



Self-stirring pot

Stirio is an expandable blade that sits on any pot and stirs on its own for a full hour on a single charge <http://dgit.in/Stirio>



3D Bionic Eye

Princeton's 3D LED printer integrates quantum dot LEDs with contact lenses to project beams of coloured light <http://dgit.in/Bionic3DEye>

From the labs

Although PDS is designed to be non-lethal, the method implemented to deliver the electric charge can be tweaked to the point that it can deliver up to 10,000 volts of electricity, enough to fry a large dog. Developed in 2007, the system is currently deployed in military buildings across the US. Fortunately, plans to make this device available for civilian use never materialized.

Personal Projectile Launcher

The ARMA 100 is probably the most easily accessible non-lethal weapon on this list. It is a handheld projectile launcher capable of providing just the right amount of nonlethal force to injure an attacker from a distance of up to 20 feet away. The idea behind using this amount of force is that it can give you enough time to escape without the need for anyone to resort to deadly force. Approximately the size of an air gun, it uses single-shot gas cartridges for propulsion and has an effective range of up to, you guessed it, 20 feet.

The gun uses a custom bean-bag-type projectile that if fired at a vulnerable spot, such as the stomach or head, can cause a good deal of damage to the target. It weighs around 1.5 lbs with a barrel length of 10 inches and requires a basic knowledge of gun handling to operate. The ARMA 100 is a relatively safer option for those inclined towards purchasing a viable self-defense device for home. Furthermore, it is legal to use without a license (in the United States) and can be purchased without a background check.

Shock and Awe

The name Taser is now synonymous with non-lethal weaponry. So much so, that any generic electric gun is simply referred to as a Taser—not unlike how all photocopiers are referred to as Xerox machines. Taser guns now come primarily in two flavours: the smaller version that most law enforcement personnel roam with is the X26P. The original model (X26) developed in 2003 featured a gas tube spark gap and fixed discharge energy, whereas the new one features the ability to keep a log of discharges, as well as an advanced digital



Law and Order: Shocking Victims Unit – “But officer, I paid for these donuts!”

display and extended battery. Then there's the X2, which is water-resistant and features dual lasers that improve accuracy.

Both these models, however, look like a child's toy in front of their (now discontinued) bigger brother, the X12. This 12-gauge shotgun fires Taser XREP cartridges (i.e., Extended Range Electronic Projectiles) up to a range of 100 feet. Like the X26P, the projectiles of X12 are “self-contained, wireless projectile(s) that delivers the same neuro-muscular incapacitation bio-effect,” which in plain English translates to “a very painful electric shock.”

Being in the weapons industry, Taser is no stranger to controversy. In fact, their products have not always proven to be “non-lethal”; some critics have claimed that Taser rounds are capable of inducing cardiac arrhythmia in targets, eventually leading to a heart attack and instant death. Every year, reports crop up listing police brutality with many incidents involving unsolicited use of taser guns. It is obvious that even though the taser gun is marketed as a non-lethal weapon, it is extremely important that its use be regulated. Proper training for its users cannot be understated.

Cloudy with a hint of chemicals

We were bound to come across a highly controversial weapon developed by

the US military sooner than later. The XM1063 is the codename for the US Army's new secret non-lethal weapon. The development of the weapon was overseen by ARDEC, the US Army's Armament Research, Development and Engineering Centre.

Details about this mysterious weapon are still rather murky. Many reports have stated that it is an artillery weapon that shoots a projectile which delivers a payload containing a concoction of chemicals devised to immobilize and incapacitate targets without killing them. The projectile is fired using a 155 mm howitzer that has a range of 28 km. As the payload parachutes down towards the designated target, it releases the toxic cocktail over an area of effectively one acre.

Nobody is still entirely sure about the exact nature of the payload, although there are numerous speculations about it. Some experts claim that it may contain a type of riot-control agent, similar to that seen in tear gas grenades and pepper sprays. Others state that it could simply be a malodorous (such as a super-powered stink bomb) or an anti-traction agent that makes the target area extremely slippery and impossible to traverse. It is possible that we'd see the application of this weapon in anti-terrorism operations to incapacitate and capture targets. 